Label collision avoidance

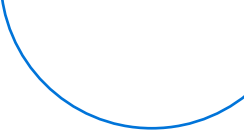
Part 1 — numeric assertions

All numeric assertions passed.

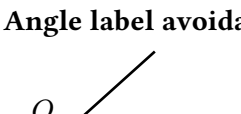
Part 2 — visual cases

Five crowded points all requesting "above" (too tight for one side → alternate above/below, full size)

Same figure fully automatic (endpoints go outward, middle row stays calm)

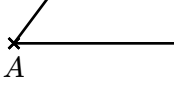


Moderate crowding stays at full size (no shrink unless nothing calm fits)

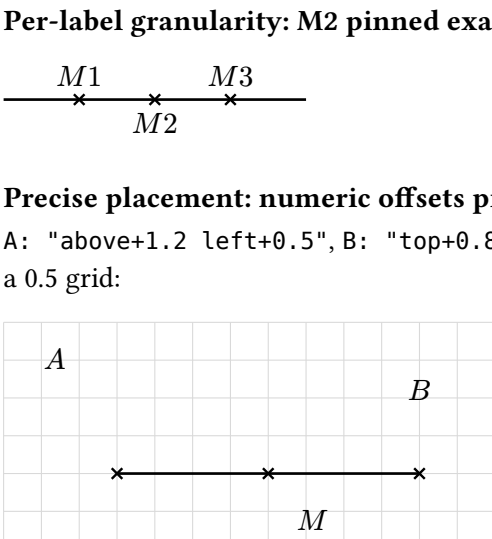


Same five crowded points with ctz-label-avoid((allow-shrink: false)) (alternation needs no shrink anyway)

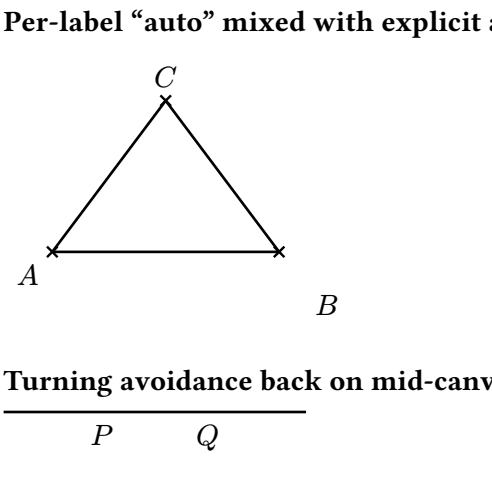
Same figure with avoidance disabled via ctz-label-avoid(false) (labels overlap — for comparison)



Crowded triangle: circumcircle + incircle + medians, everything labeled



Same triangle, fully automatic (labels: true — no positions specified)



Point on a line, label "below" blocked by a second line → flips to the open side above

Scale independence: same figure at length 0.5cm and 2cm

Labels avoid a circle stroke but are allowed inside the circle

Angle label avoidance: α slides past the Q marker; Q's label dodges both

Duplicate labels are skipped (path auto-labels + explicit ctz-draw-labels)

Per-label granularity: M2 pinned exactly where asked (avoid: false), M1/M3 dodge

Precise placement: numeric offsets pin labels exactly (engine off for them)

`A: "above+1.2 left+0.5"`, `B: "top+0.8"`, `M: "below+0.3 right+0.2"` — offsets in canvas units on a 0.5 grid:

Per-label "auto" mixed with explicit and precise positions

Turning avoidance back on mid-canvas